

ET.PM4 Firmware Subsystem Test

The *Firmware Subsystem Test* is an intermediate milestone in the ET.PM4 project module. Its primary purpose is to verify that the A/D conversion, the digital signal acquisition and the (complex-) FFT spectrum calculation implemented by the project groups operate as intended and provide a fully functional and a stable foundation for the other system components and for the complete device.

The ET.PM4 project groups shall present either on paper or on a PC display:

- Documented tests and their results.
- A block diagram showing the implemented and planned functional digital signal processing blocks required to achieve the intended final device functionality.
- Explain the principles of the intended user interface of the device e.g., using sketches of example screen shots.

In addition, they shall demonstrate live the correct operation of the implemented digital signal acquisition and the FFT with an FFT spectrum displayed on the microcontroller board's LCD.

The specific goals are to show that:

- The displayed FFT spectrum shows the expected frequency.
- The displayed signal spectrum is of good quality, i.e. that it is stable and that noise and other possible interferences would not significantly affect intended device functionality.

Presentation and demonstration sessions:

- May typically last 5 minutes but not longer than approximately 10 minutes.
- Will take place in the classroom at the student's desks.
- All members of the project groups must be present.
- Should preferably be performed immediately after the Hardware Subsystem Test presentation and demonstration.
- Audience: the technical ET.PM4 lecturer

Tests of individual firmware features:

Not required, but may optionally be presented and / or demonstrated.

Grading

Based on this presentation and demonstration, the technical lecturer will determine a partial grade, which contributes 15% to the overall ET.PM4 course grade.

See also the following documents on the ET.PM4 pool:

- Checklist_Marks_FS2026.xlsx
- PM4_FS26_Subsystem_Test_Plan.pdf